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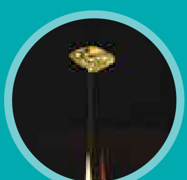
SCIENCE IS THE PATH TO KNOWLEDGE, AND FINDING OUT ABOUT HOW OUR UNIVERSE WORKS. YOU CAN'T BE A GEEK AND NOT KNOW YOUR SCIENCE!

THIS MONTH IN SCIENCE:

We explore some mystical twin towns around the world, delve into possible explanations for why we sleep, and discuss what exactly went wrong with the Chandrayaan 2 mission.



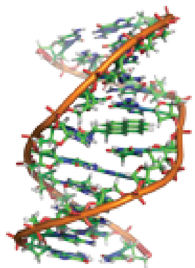
The blackest black

The blackest material known, made from carbon nanotubes is about ten times darker than any other known superblack materials, and yes that includes Vantablack. The material absorbed 99.995 of all incident light, and any bumps or features on the material are invisible no matter what angle they are seen from. **https://dgit.in/blkstblk**

WHAT'S NEW

Double helix structure may have emerged with ease

One of the mysteries of the origin of life, is how the double helix structure of DNA and RNA first appeared. According to a study, this enigma is not hard to explain and that RNA's chemical ancestors may have casually started coiling up into spiral strands, billions of years ago. In the lab, it is a very difficult task to make RNA or DNA chains without using enzymes, which act as a tool for assembling and replicating strands. The researchers instead used proto-nucleobases, molecules that could be the ancestors of the nucleobases in use within RNA and DNA today. They found that the polymers resembling RNA



acquired the spiral shape on their own, in water, without the use of an enzyme, at room temperatures. Scientists believe that these polymers referred to as "assemblies" acquired the spiral shape, evolved into RNA and then DNA. A major difference between the assemblies and RNA today is that proto nucleobases in the assemblies had non covalent bonds while the nucleobases in RNA form covalent bonds. **https://dgit.in/ezrna**

Water discovered on habitable super-earth

For the first time, astronomers have detected the presence of water in an exoplanet orbiting in the habitable zone around its host star. K2-18b is a super Earth, 124 light years away towards the constellation of Leo. Using data from the Kepler and Hubble space telescopes, a series of transits were observed, where the planet passed in front of the host star in relation to the Earth. The light from these transits was analysed by open source algo-

rithms, to find the telltale signs of chemicals. Apart from water vapour, the findings also indicated the presence of hydrogen and helium. However, there are some problems concerning the habitability of the planet. The red dwarf star that it orbits around may be too active to support life. The planet also has gravity several times that of Earth, and may actually be closer to Neptune. The planet is enveloped in a thick, heavy atmosphere, which may not be conducive to life. K2-18b is about ten times as massive as the Earth. However, the planet remains a promising candidate for observations and potentially finding biosignatures. **https://dgit.in/sewater**

Mummy health checkup

University of Manchester received two Egyptian mummies for a health checkup of sorts. The mummies were photographed, x-rayed and CT-scanned. The imaging techniques allowed researchers to identify the sexes of the two mummies, as well as estimate their social status. **https://dgit.in/mmychkup**

Pearl-inspired armour

Researchers imitated the outer coating on pearls, known as nacre on plastics, and discovered that it is 14 times stronger than steel, and ideal for absorbing the impacts of projectiles. Researchers are now looking for ways to make body armour from the material. **https://dgit.in/prlarmr**

Largest optical lens

One of the primary components of the Large Synoptic Survey Telescope (LSST) is the lens assembly. There are three lenses, two of which are 1.2m and 1.57m in diameter. The 1.57m lens supposedly has the largest optical lens ever fabricated. The assembly took over five years to realise. **https://dgit.in/olens**